Fiscal Year:	FY 2023	**Task Last Updated:**	FY 02/10/2023
PI Name:	Naithani, Sushma Ph.D.		
Project Title:	In Silico Modeling of Gene Networks and Mechanisms Associated With Plant Gravitropism		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	97333	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2020 Space Biology NNH20ZDA001N-SB E.12. Flight/Ground Research
Start Date:	04/06/2022	**End Date:**	04/05/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA ARC
Contact Monitor:	Zhang, Ye	**Contact Phone:**	321-861-3253
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 04/05/2024 per NSSC information (Ed., 10/30/23).		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):			
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Task Description:	Gravity acts as an important determinant for shaping a plant's architecture. Understanding plant growth, development, and survival in response to altered/microgravity is crucial for developing sustainable bioregenerative life support systems for astronauts in long-term spaceflights, the International Space Station (ISS), and colonization of the Moon and Mars. The available research data suggest that plants can grow and complete their life cycles in the microgravity environment of spaceflights and the ISS; however, altered gravity and light conditions impact plant productivity, vigor, and starch storage organs (e.g., seeds and tubers). Furthermore, several independent studies on Arabidopsis (a few on rice and other species) mutants defective in hormone signaling and gravitropic response have contributed to the identification of the molecular components and mechanisms associated with plant gravitropism and how it is influenced by other tropism associated with environmental stimuli such as light, availability of water, and nutrients. In recent years, a few transcriptomic, proteomic, and metabolomic studies on plants subjected to altered gravity in spaceflights and on mutants defective in gravitropism have identified differential expression of genes associated with the fundamental biological processes of photosynthesis, phototropism, gas exchange, plant anchoring, pigment synthesis, and water and nutrient uptake. However, the lack of a system-level pathway network model for visualizing gravitropism (gravity-induced differential growth processes) and its connections with other tropisms hinders full utilization/analysis of the high-throughput transcriptomic and proteomic data. We propose to biocurate and build an in silico model of plant gravitropism response by analyzing the Omics datasets from Arabidopsis thaliana available at NASA GeneLab; https://) and based on a review of published literature. These pathways will be used for deriving the projection of pathways for an additional 106 species, ranging from unicellular photoautotrophs to higher plants, including several models and crop plants. The curated pathways and gene-orthology-based pathways will be hosted on the Plant Reactome knowledgebase (https://), an open-source pathway portal of the Gramene project. Plant Reactome can (i) serve as a nodal point for knowledge synthesis about gravitropism and associated genes, regulatory networks, and molecular mechanisms within the context of overall plant development and metabolism; (ii) provide a platform to space biology researchers in visualizing and analyzing Omics datasets; (iii) extend the knowledge of genes and pathways that influence gravity response or intersect with other regulatory mechanisms (phototropism, chemotropism, hydrotropism, and hormone signaling); and (iv) inform about evolutionary relevance of the conclusions derived from the Arabidopsis. Overall, our proposal will support space biology researchers to formulate data-driven hypotheses for designing plants suited for cultivation in controlled environments in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our project involves biocuration of pathways and gene-regulatory networks involved in plant gravitropism within the Plant Reactome Knowledgebase that serves as a free-of-cost, public resource for plant genomics scientists, breeders, and educators. In particular, in silico modeling of gene-gene networks and gravitropism associated pathways provide a conceptual framework for understanding how plant development and architecture is shaped by interaction of gravity with the plants overall genetic developmental program and other environmental stimuli. It is well aligned with NASA's space biology goals "Create new knowledge of how biological systems adapt to space that can translate into benefits for NASA Exploration Missions and for Earth applications through sponsoring competitively solicited Space Biology research" and "Train and inspire a new generation of U.S. Space Biologists". It can (i) serve as a nodal point for the synthesis of knowledge about gravitropism; (ii) provide a platform to space biology researchers to upload, visualize, and analyze Omics datasets of their choice in the context of plant metabolic and transport pathways, gene-networks, and plant developmental processes; (iii) extend the knowledge of genes and pathways that influence gravity response or intersect with other regulatory mechanisms (phototropism; chemotropism, and hormone signaling); and (iv) inform about evolutionary relevance of the conclusion derived from the Arabidopsis and rice (reference species in the Plant Reactome). Overall, our proposal support space biology researchers to formulate data-driven hypotheses for designing plants suited for cultivation in controlled environments in space flights, ISS, and beyond.
	In silico modeling of gene networks and mechanisms associated with plant gravitropism Progress report 4/6/22 – 2/10/23 (We received funding confirmation on 06/26/2022 and started working in October 2022). Principal investigator: Sushma Naithani Department of Botany and Plant Pathology Cordley Hall, 2701 SW Campus Way, Oregon State University Corvallis, OR 97331, USA Grant/Contract number: 80NSSC22K0891 1. Abstract: This project aims to (i) biocurate the in silico model of plant gravitropism response by analyzing the Omics datasets from Arabidopsis thaliana subjected to altered gravity during spaceflights (available at NASA GeneLab; https:// and from published studies on Arabidopsis and rice mutants defective in gravitropism; (ii) run an automated pipeline that uses protein sequence conservation between plant species to infer versions of these pathways for 106 additional species and integrate all of these pathways into Plant Reactome (https:// ; and (iii) Embed a DiagramJs pathway widget (a dynamic, interactive pathway viewer), into the NASA GeneLab Data Systems (GLDS) website to support easy navigation of the Plant Reactome database by space biology researchers. Based on Omics data analysis of the datasets available in the GeneLab, we have selected 500 unique Arabidopsis genes that show differential gene expression or have a mutant defective in gravitropism. These selected genes include transcription factors, transporters, kinases, and metabolic enzymes associated with gravitropism and do not currently exist (projected in Arabidopsis pathways) in the Plant Reactome. Since we use rice as a reference species for biocuration of pathways in the Plant Reactome knowledgebase, we have identified corresponding orthologs from rice (Oryza sativa). A careful comparison between Arabidopsis and rice will help to identify the conserved as well as diversified gene functions and molecular mechanisms associated with plant gravitropism in both dicots and monocots, as well as improve the further automated projections of the pathways on other species. To connect the OMICs data analysis results with published experimental evidence on gene function and gene-gene interactions, we have selected 50 research and review articles from dicot model species Arabidopsis and from monocot model species rice (Oryza sativa). We are working with five undergraduate students funded by Undergraduate Research, Scholarships provided by Oregon State University. The undergraduate students are being trained for extraction of the information about genes from the published scientific papers. The critical review of the scientific literature is an

Task Progress:	essential step for extracting the evidence related to the gene function, mutant phenotypes, and gene-gene interactions involved in the gravitropism at various stages of the plant's life cycle (i.e., seed germination and early seedling development, shoot growth and branching, leaf angle formation, root growth and differentiation). This basic organization of information helps us to create a framework for analyzing how gravity is sensed, to map the major transcriptional events associated with signal transduction in response to gravity, linking phytohormone signaling, and the differential growth phenotypes and traits. The undergraduate biocuration training supports our current project. Secondly, we established a standard protocol for using transcriptome data for biocuration of transcription factors (TF) networks. A TF network was constructed based on the two lines of evidence: co-expression correlation of differentially expressed genes, and the presence of Transcriptional Factor binding site(s) in the target gene's promoter. In addition, we extracted the experimental evidence on gene function and gene-gene interaction to support the gene-gene interactions and added additional genes in this network. Finally, we construct gene-gene interaction maps by manually curating the reactions and pathways using the Reactome Curator Tool. We have successfully biocurated a pathway category, "Gravitropism under normal or artificial gravity environment", and have submitted our first sub-pathway, "Regulation of seed germination and coleoptile growth under submergence and normal gravity environment". The pathway will be available in the public domain after updated release of the Plant Reactome. Besides using the genes extracted from GeneLab transcriptome data, we added genes in this network from the gravitropism defective mutants of Arabidopsis or rice from published literature, and a few genes mined from a study involving analysis of submergence tolerance in various rice genotypes during seed germination. Unlike Arabidopsis, the cultivated rice O. sativa and its wild ancestors have varying degrees of submergence tolerance; therefore, including these genes provided insights into interactions among the genes involved in genetic regulation of seed germination and coleoptile growth, submergence and hypoxia tolerance, and differential growth in response to gravity. Interestingly, plants grown under a microgravity environment in spaceflights and at space stations experience reduced bioavailability of oxygen or hypoxia stress. On Earth’s environment, plants experience hypoxia stress during submergence. Thus, the synthesis of the transcriptional network by integrating heterogenous data and information can be achieved using the Plant Reactome platform, which in turn can help the orthology-based pathway projections to a wide range of species, including both dicots and monocots. We are implementing a similar strategy to identify gene-gene networks and biocuration of pathways involved in shoot growth and branching, formation of leaf angle, and root gravitropism by analyzing the Omics data from Arabidopsis from the NASA GeneLab and based on additional information from literature. Wherever possible, we are using additional transcriptome data from rice to enrich our pathway curations. 2. Research products: i. Developed a conceptual framework for organizing plant gravitropism pathways in the Plant Reactome. A transcription factors network consisting of ~60 genes was biocurated and submitted to the Reactome central database. ii. Established a standard protocol for mining gene-gene interaction using transcriptome data. iii. Preliminary review of 50 scientific research papers from rice and Arabidopsis has been conducted and a list of 500 important Arabidopsis genes associated with plant gravitropism has been compiled and mapped to rice orthologues. iv. Training five undergraduate students in genomic data literacy and biocuration of genes and pathways. 3. Request for no cost extension: We are requesting a no-cost extension for one year at the end of the project. The funding for this project was received in the last week of June 2022; that gives us only 9 months’ time instead of 1 year as proposed by us. In addition, the start of the project was further delayed due to the relocation of our laboratory and office from 4575 SW Research Way to 2701 SW Campus Way in July-August of 2022. We could only start working on this project in October 2022 due to relocation of the lab and office. Goals for next year: We aim to complete the project by 04/06/2024. This will give us sufficient time to meet the following objectives. i. Complete the integration of all 500 gravitropism associated Arabidopsis genes (~ 60 genes added as of Feb 2023) and their rice orthologs in pathways and processes involved in shoot growth and branching, root growth and differentiation, leaf angle formation. ii. Run an automated pipeline that uses protein sequence conservation between plant species to infer versions of these pathways for 120 additional species and integrate all of these pathways into the Plant Reactome knowledgebase. iii. Embed a DiagramJs pathway widget (a dynamic, interactive pathway viewer) into the GLDS website to support easy navigation of the Plant Reactome database by space biology researchers. 4. Budget summary: The work and the budget for the next year of the effort will be consistent with the work plan and budget previously approved by NASA. No changes are requested for the final year budget for this project. We are requesting a no-cost extension for one year at the end of the project. The funding for this project was received in June/26/2022 and the start of the project was delayed until October 2022 due to the relocation of our laboratory and office from 4575 SW Research Way to 2701 SW Campus Way in July-August of 2022. 5. Publications and presentations: see Cumulative Bibliography. 6. Training: The project is training five undergraduate students on genomic data literacy and biocuration.
Bibliography Type:	Description: (Last Updated: 02/12/2026)
Abstracts for Journals and Proceedings	Naithani S. "Using OMICs data for biocuration of plant genes and pathways for the Plant Reactome." International Plant and Animal Genome Conference / PAG 30, San Diego, CA, January 13-18, 2023. Abstracts. International Plant and Animal Genome Conference / PAG 30, San Diego, CA, January 13-18, 2023. , Jan-2023
Abstracts for Journals and Proceedings	Naithani S. "Plant Reactome: Using OMICs data for biocuration of plant genes and pathways." AgBioData Webinar Series, Virtual, December 7, 2022. Abstracts. AgBioData Webinar Series, Virtual, December 7, 2022. https://www.agbiodata.org/meetings , Dec-2022
Abstracts for Journals and Proceedings	Naithani S. "Plant Reactome resource for space biologists." AWG Symposium 2022 (NASA GeneLab Analysis Working Group Symposium), Virtual, June 30, 2022. Abstracts. AWG Symposium 2022 (NASA GeneLab Analysis Working Group Symposium), Virtual, June 30, 2022. , Jun-2022

Abstracts for Journals and Proceedings	Naithani S, El-Husseini S, Hooks E, Elser J, Mohanty B, Olson A, Wei S, D'Eustachio P, Ware D, Jaiswal P. "Plant Reactome Knowledgebase: Using OMICs sata for biocuration of plant genes and pathways." International Plant and Animal Genome Conference / PAG 30, San Diego, CA, January 13-18, 2023. Abstracts. International Plant and Animal Genome Conference / PAG 30, San Diego, CA, January 13-18, 2023. , Jan-2023